



Hotel Revenue Analysis Project

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Link to GitHub Repository Page:

[Github Repository Page](#)

Problem Statement

- Goal: A business manager of a chain of hotel would like to request help from the data analytics team to develop a dashboard to analyze & visualize hotel booking data
- The business manager is interested if there's any patterns/ seasonality with respect to:
 1. Guests
 2. Revenue

Requirement: Build a Dashboard using PowerBI to provide at-a-glance information about features that are relevant to revenue

Dataset Background

- The dataset used in this analysis can be found in my GitHub

Repository:

- [Github Repository Page](#)
- This data set contains booking information for a city hotel and a resort hotel
- In the following slide, further elaboration into the fieldname, as well as the definition of each field will be shown.

Dataset glossary in final SQL query [1/4]

Original Columns Present in Dataset

Field Name	Description
Hotel	<i>Hotel (H1 = Resort Hotel or H2 = City Hotel)</i>
is_canceled	<i>Value indicating if the booking was canceled (1) or not (0)</i>
lead_time	<i>Number of days that elapsed between the entering date of the booking into the PMS and the arrival date</i>
Arrival_date_year	<i>Year of arrival date</i>
Arrival_date_month	<i>Month of arrival date</i>
Arrival_date_week_number	<i>Week number of year for arrival date</i>
Arrival_date_day_of_month	<i>Day of arrival date</i>
Stays_in_weekend_nights	<i>Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel</i>
Stays_in_week_nights	<i>Number of week nights (Monday to Friday) the guest stayed or booked to stay at the hotel</i>
Adults	<i>Number of adults</i>
Children	<i>Number of children</i>
Babies	<i>Number of babies</i>
Cost	<i>Cost incurred by the hotel company related to the associated meal package provided to the customer</i>
Meal	<i>Type of meal booked. Categories are presented in standard hospitality meal packages: Undefined/SC – no meal package; BB – Bed & Breakfast; HB – Half board (breakfast and one other meal – usually dinner); FB – Full board (breakfast, lunch and dinner)</i>
country	<i>Country of origin. Categories are represented in the ISO 3155–3:2013 format</i>

Dataset glossary in final SQL query [2/4]

Original Columns Present in Dataset

Field Name	Description
Market_segment	Market segment designation. In categories, the term “TA” means “Travel Agents” and “TO” means “Tour Operators”
Distribution_channel	Booking distribution channel. The term “TA” means “Travel Agents” and “TO” means “Tour Operators”
Is_repeated_guest	Value indicating if the booking name was from a repeated guest (1) or not (0)
Previous_cancellations	Number of previous bookings that were cancelled by the customer prior to the current booking
Previous_bookings_not_cancelled	Number of previous bookings not cancelled by the customer prior to the current booking
Reserved_room_type	Code of room type reserved. Code is presented instead of designation for anonymity reasons.
Assigned_room_type	Code for the type of room assigned to the booking. Sometimes the assigned room type differs from the reserved room type due to hotel operation reasons (e.g. overbooking) or by customer request. Code is presented instead of designation for anonymity reasons.
Booking_changes	Number of changes/amendments made to the booking from the moment the booking was entered on the PMS until the moment of check-in or cancellation
Deposit_type	Indication on if the customer made a deposit to guarantee the booking. This variable can assume three categories: No Deposit – no deposit was made; Non Refund – a deposit was made in the value of the total stay cost; Refundable – a deposit was made with a value under the total cost of stay.
Agent	ID of the travel agency that made the booking
Company	ID of the company/entity that made the booking or responsible for paying the booking. ID is presented instead of designation for anonymity reasons

Dataset glossary in final SQL query [3/4]

Original Columns Present in Dataset

Field Name	Description
days_in_waiting_list	Number of days the booking was in the waiting list before it was confirmed to the customer
Customer_type	Type of booking, assuming one of four categories: Contract - when the booking has an allotment or other type of contract associated to it; Group - when the booking is associated to a group; Transient - when the booking is not part of a group or contract, and is not associated to other transient booking; Transient-party - when the booking is transient, but is associated to at least one other transient booking
Adr	Average Daily Rate as defined by dividing the sum of all lodging transactions by the total number of staying nights
Required_car_parking_spaces	Number of car parking spaces required by the customer
total_of_special_requests	Number of special requests made by the customer (e.g. twin bed or high floor)
reservation_status	Reservation last status, assuming one of three categories: Canceled - booking was canceled by the customer; Check-Out - customer has checked in but already departed; No-Show - customer did not check-in and did not inform the hotel of the reason why
Reservation_status_date	Date at which the last status was set. This variable can be used in conjunction with the ReservationStatus to understand when the booking was canceled or when the customer checked-out of the hotel
Discount	Float value ranging from 0 to 1. Contains the associated discount provided to the customer belonging to the market_segment
Market_segment	Type of market_segment that the customer belongs to. Can be classified as: Undefined, Direct, Groups, Corporate, Aviation, Offline TA/TO, Online TA, Complementary

Dataset glossary in final SQL query [4/4]

Self-Calculated Columns

Field Name	Description
Revenue	<i>Self-Calculated column based on the factors:</i> • <i>Stays_in_week_nights</i> • <i>Stays_in_weekend_nights</i> • <i>ADR</i> • <i>Cost</i> • <i>Discount</i>
Guest_type	<i>Self-Calculated column based on the factors:</i> • <i># of adults</i> • <i># of children</i> • <i># of babies</i> <i>To categorize guest type into 4 categories: Single, Couple, Group, Family</i>
Region	<i>Self-Calculated column based on the guest's country code, which is represented in the ISO 3155–3:2013 format</i>
Seasons	<i>Self-Calculated column based on the 'arrival_date_month' of guests</i>
Day_of_week	<i>Self-Calculated column based on the 'arrival_date_day_of_month', 'arrival_date_month', 'arrival_date_year' of guests</i>

Dataset Preparation in SQL [1/7]

Creation of Baseline Query

- Raw .csvfile is ingested into Microsoft SQL Server Management Studio (SSMS) and SQL queries were iteratively built upon the baseline query below to obtain the final query to import into SSMS
- The baseline query utilizes UNION ALL, CTE & LEFT JOINS to aggregate all sheets within the excel spreadsheet

```
/*
  Baseline Query to consolidate data across all tables
*/
-- Given that we'd like to analyze trends, revenue growth across each year, and that the names of
columns remain the same across each year. UNION ALL was performed across all years to consolidate
data. We'll use CTE and refer to the consolidated table as Hotel. At the same time, the foreign keys
market_segment, meal was used to JOIN market_segment$ & meal_cost$ tables together:
WITH hotels AS (
  SELECT * FROM Hotel_Revenue_Dataset.dbo.['2018$']
  UNION ALL
  SELECT * FROM Hotel_Revenue_Dataset.dbo.['2019$']
  UNION ALL
  SELECT * FROM Hotel_Revenue_Dataset.dbo.['2020$']
)
SELECT *
FROM hotels as h
LEFT JOIN Hotel_Revenue_Dataset.dbo.market_segment$ AS m ON h.market_segment = m.market_segment
LEFT JOIN Hotel_Revenue_Dataset.dbo.meal_cost$ AS c ON h.meal = c.meal
```

- Following which, SQL queries were iteratively built upon for the creation of revenue & categorical columns

Dataset Preparation in SQL [2/7]

Creation of 'guest_type' column

One of the requirements was to investigate if there's any guest-related seasonality or trends

Noticing that in the dataset, there are 'adults', 'children' & 'babies' information for each booking made

- I want to create a single column that categorizes each booking by the type of guests, they fall into 4 categories
 - Single
 - Couple
 - Group
 - Family

```
-- Categorical Query 1: creation of guest_type categorization code, we want to categorize guest_type based on adults,
children, babies variable
WITH hotels AS (
  SELECT * FROM Hotel_Revenue_Dataset.dbo.[2018$]
  UNION ALL
  SELECT * FROM Hotel_Revenue_Dataset.dbo.[2019$]
  UNION ALL
  SELECT * FROM Hotel_Revenue_Dataset.dbo.[2020$]
),
hotel_all_columns_with_guest_type AS (
  SELECT adults, children, babies,
    CASE
      WHEN adults = 1 AND children = 0 AND babies = 0 THEN 'Single'
      WHEN adults = 2 AND children = 0 AND babies = 0 THEN 'Couple'
      WHEN adults > 2 AND children = 0 AND babies = 0 THEN 'Group'
      ELSE 'Family'
    END AS guest_type
  FROM hotels AS h
  LEFT JOIN Hotel_Revenue_Dataset.dbo.market_segment$ AS m ON h.market_segment = m.market_segment
  LEFT JOIN Hotel_Revenue_Dataset.dbo.meal_cost$ AS c ON h.meal = c.meal
)
SELECT *
FROM hotel_all_columns_with_guest_type
```

Dataset Preparation in SQL [3/7]

Creation of 'region' column

Next, I noticed that the countries in which the guests belong to are stored in a three letter country code format.

I'd like to create a categorical column called 'region' to see if we can get any insights related to the data with regards to the customer's region

- The somewhat lengthy SQL code below utilizes multiple CASE WHEN and IN statements to return the corresponding regions of the guests

[illegible]

Dataset Preparation in SQL [4/7]

Creation of 'seasons' column

Given that the month of arrival of guests are provided in the dataset, I'd like to categorize them by seasons:

```
-- Categorical Query 3): creation of seasons categorization code:
WITH hotels AS (
    SELECT * FROM Hotel_Revenue_Dataset.dbo.['2018$']
    UNION ALL
    SELECT * FROM Hotel_Revenue_Dataset.dbo.['2019$']
    UNION ALL
    SELECT * FROM Hotel_Revenue_Dataset.dbo.['2020$']
),
hotel_all_columns_with_seasons AS (
    SELECT arrival_date_month,
           CASE
               WHEN arrival_date_month IN ('December', 'January', 'February') THEN 'Winter'
               WHEN arrival_date_month IN ('March', 'April', 'May') THEN 'Spring'
               WHEN arrival_date_month IN ('June', 'July', 'August') THEN 'Summer'
               WHEN arrival_date_month IN ('September', 'October', 'November') THEN 'Autumn'
           END AS seasons
    FROM hotels AS h
    LEFT JOIN Hotel_Revenue_Dataset.dbo.market_segment$ AS m ON h.market_segment = m.market_segment
    LEFT JOIN Hotel_Revenue_Dataset.dbo.meal_cost$ AS c ON h.meal = c.meal
)
SELECT *
FROM hotel_all_columns_with_seasons
```

Dataset Preparation in SQL [5/7]

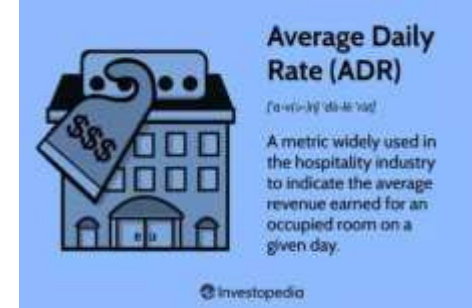
Creation of 'day_of_week' column

- The SQL query below utilizes DATEPART & CASE WHEN statements, as well as converting and concatenating each column from the arrival date to extract out the corresponding day of the week in which the guests arrive in.

```
-- Categorical Query 4): creation of day_of_week categorization code
WITH hotels AS (
  SELECT * FROM Hotel_Revenue_Dataset.dbo.[ '2018$' ]
  UNION ALL
  SELECT * FROM Hotel_Revenue_Dataset.dbo.[ '2019$' ]
  UNION ALL
  SELECT * FROM Hotel_Revenue_Dataset.dbo.[ '2020$' ]
)
SELECT
  arrival_date_day_of_month,
  arrival_date_month,
  arrival_date_year,
  CONVERT(DATE, CONVERT(VARCHAR, arrival_date_day_of_month) + ' ' + arrival_date_month + ' ' +
  CONVERT(VARCHAR, arrival_date_year)) AS arrival_date, -- concat and convert year, month, day of month
  -- into date format. We name it as arrival_date column.
  CASE DATEPART(dw, CONVERT(DATE, CONVERT(VARCHAR, arrival_date_day_of_month) + ' ' +
  arrival_date_month + ' ' + CONVERT(VARCHAR, arrival_date_year))) -- DATEPART(dw, ...) extracts day
  of the week as an integer based on the CASE statement. CASE statement then converts integer to it's
  corresponding day name. (I.e. 'Sunday' is dw value corresponding to 1)
  WHEN 1 THEN 'Sunday'
  WHEN 2 THEN 'Monday'
  WHEN 3 THEN 'Tuesday'
  WHEN 4 THEN 'Wednesday'
  WHEN 5 THEN 'Thursday'
  WHEN 6 THEN 'Friday'
  WHEN 7 THEN 'Saturday'
  END AS day_of_week
FROM hotels;
```


Dataset Preparation in SQL [6/7]

Creation of 'revenue' column



```
-- Categorical Query 5): creation of revenue categorization code:
WITH hotels AS (
  SELECT * FROM Hotel_Revenue_Dataset.dbo.[ '2018$' ]
  UNION ALL
  SELECT * FROM Hotel_Revenue_Dataset.dbo.[ '2019$' ]
  UNION ALL
  SELECT * FROM Hotel_Revenue_Dataset.dbo.[ '2020$' ]
),
hotel_all_columns_with_revenue AS (
  SELECT reservation_status, adr, discount, c.Cost,
    (stays_in_week_nights + stays_in_weekend_nights) AS Number_of_Rooms_Sold, -- used to calculate
    revenue
    CASE
      WHEN reservation_status = 'Canceled' THEN 0 -- Validated it's 0. Here we assume that
        customers who canceled the reservation did so within the full return window, hence no revenue is
        generated from such transactions.
      WHEN ADR < 0 THEN 0 -- Validated it's 0. As concluded earlier, for ADR < 0, it's
        highly likely a calculation error/ data discrepancy has occurred.
      WHEN (reservation_status = 'Check-Out' OR reservation_status = 'No-Show') AND (adr =
        0 OR adr < 0) THEN 0 -- Validated it's 0. If a customer has checked out or no show, ADR should not
        be less than or equal to 0 as the customer should be billed for such cases and thus will generate
        revenue for the company.
      ELSE
        ( ((stays_in_week_nights + stays_in_weekend_nights) * (adr) * (1 - Discount)) -
        c.Cost )
    END AS revenue
  FROM hotels AS h
  LEFT JOIN Hotel_Revenue_Dataset.dbo.market_segment$ AS m ON h.market_segment = m.market_segment
  LEFT JOIN Hotel_Revenue_Dataset.dbo.meal_cost$ AS c ON h.meal = c.meal
)
SELECT *
FROM hotel_all_columns_with_revenue
```

Since we're given the:

1. ADR value
2. # of weekdays and weekends stayed from a particular booking
3. % Discount corresponding for guests belong to a particular market_segment
4. Meal Cost of food provided to the guest

We can manipulate the ADR formula to obtain the corresponding revenue generated by each booking, this is given by:

Revenue per booking

$$= [(\text{stays in week nights} + \text{stays in weekend nights}) \times \text{ADR} \times (1 - \text{Discount})] - \text{Meal Cost}$$

Assumptions:

1. We assume that customers who canceled booking did so within the full return window
2. Customers that have made the booking, did not cancel in time, or show up on the appointment date, will still be charged and thus generate revenue

Dataset Preparation in SQL [7/7]

Amalgating all previous queries into a single giant query in SSMS to export into PowerBI

After validating the calculations of each individual query created, we now combine all queries into a single giant query to bring in the necessary columns and features for data analysis and visualization

A screenshot of a SQL query editor window with a dark background. The query is a complex SQL statement, likely a SELECT query, involving multiple table joins and calculations. The text is color-coded: keywords like SELECT, FROM, JOIN, and WHERE are in blue; table and column names are in white; and numerical values or literals are in yellow. The query is quite long and dense, filling most of the window. The window title bar at the top shows standard OS window controls (minimize, maximize, close) and a file path. The status bar at the bottom shows the current line and column position.

PowerBI Data Pre-Processing [1/3]

Combining “arrival_date_year”, “arrival_date_month”, “arrival_date_day_of_month” into a single column for continuous time axis plotting with hierarchical sorting

The dataset ported into PowerBI has separate columns for arrival information for guests, given by:

- “arrival_date_year”,
- “arrival_date_month”,
- “arrival_date_day_of_month”

This makes it difficult to hierarchically display arrival date information of guests in a continuous time axis



Aggregate Functions are applied by default

Year	Sum of arrival_date_year	Sum of required_car_parking_spaces	Parking Percentage
2014	730697	0.00	0.00%
2018	30534550	1,303.00	0.92%
2019	161649162	5,238.00	3.69%
2020	73695267	2,270.00	1.60%
Total	286609684	8,819.00	6.21%

We don't want this
We don't want an aggregate of the year

A custom column named “arrival_DD_MM_YYYY” was created in PowerBI with the formula below:

```
Text.From([arrival_date_day_of_month]) & "/" &  
Text.From([arrival_date_month]) & "/" &  
Text.From([arrival_date_year])
```

The new column was converted to DateTime Format:

arrival_date_day_of_month	arrival_date_month	arrival_date_year	arrival_DD_MM_YYYY
1	July	2018	1/7/2018 12:00:00 am
1	July	2018	1/7/2018 12:00:00 am
1	July	2018	1/7/2018 12:00:00 am
1	July	2018	1/7/2018 12:00:00 am
1	July	2018	1/7/2018 12:00:00 am
2	July	2018	2/7/2018 12:00:00 am
2	July	2018	2/7/2018 12:00:00 am

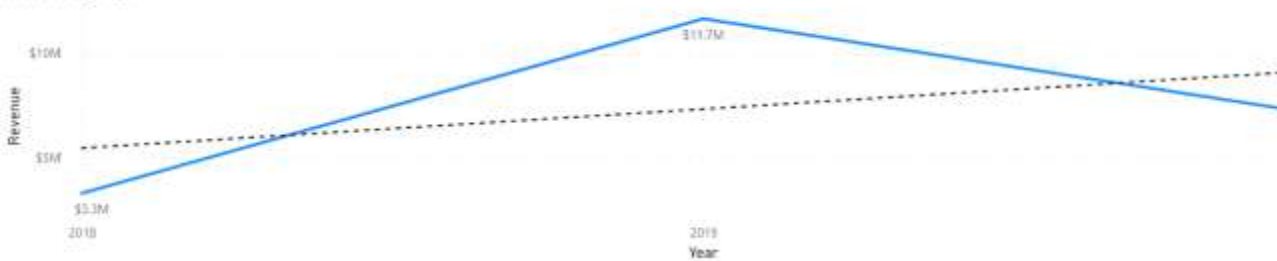
DD/MM/YYYY Format

PowerBI Data Pre-Processing [2/3]

Combining “arrival_date_year”, “arrival_date_month”, “arrival_date_day_of_month” into a single column for continuous time axis plotting with hierarchical sorting

- With the new “arrival_DD_MM_YYYY” column, we can finally display guests’ arrival time & date details in a hierarchical format:
 - By Year
 - By Quarter
 - By Month
 - By Day

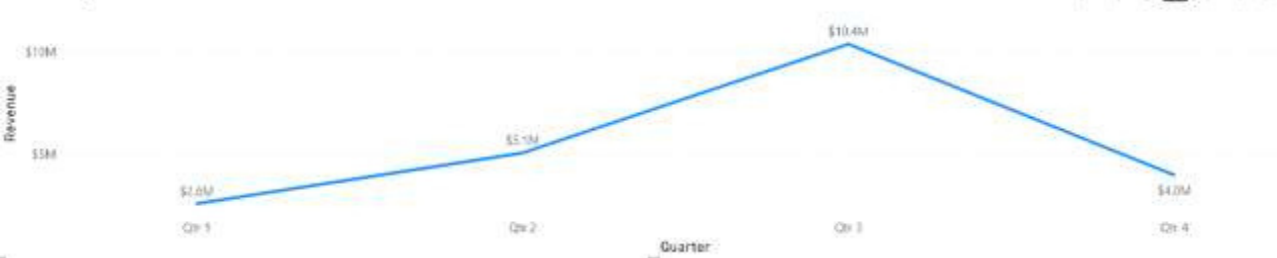
Revenue by Year



Revenue by Month



Revenue by Quarter



Revenue by Day



PowerBI Data Pre-Processing [3/3]

Creating 'WeekdayNumericTable' to sort guests booking by day_of_week

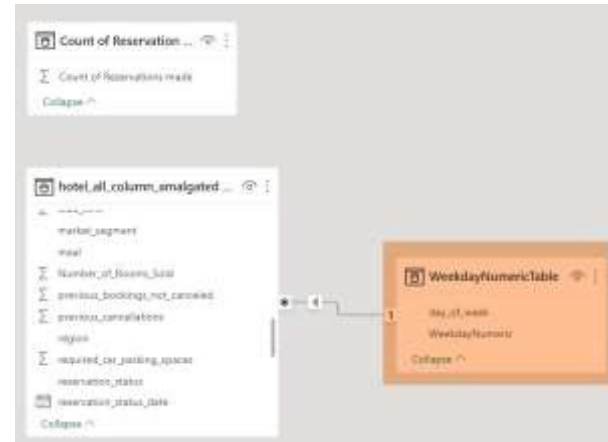
PowerBI sorts days of the week in alphabetical order by default.

- To logically sort 'day_of_week', we created a table with the same column name 'day_of_week'
- Now, 'day_of_week' from the original query is able to act as a foreign key and establish a relationship with 'day_of_week'

Create Table

The first row of data that you pasted has been promoted to a header row.

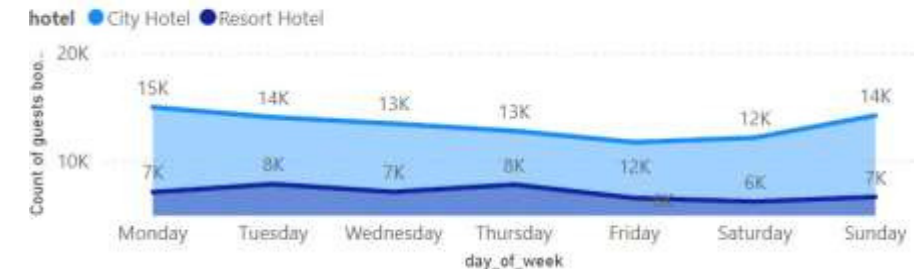
	day_of_week	WeekdayNumeric
1	Monday	1
2	Tuesday	2
3	Wednesday	3
4	Thursday	4
5	Friday	5
6	Saturday	6
7	Sunday	7



- Finally, 'day_of_week' was assigned to sort by column, using the 'WeekdayNumeric' column, to obtain the weekday-sorted plot

The screenshot shows the Power BI interface with the 'day_of_week' column selected. The 'Sort by column' dropdown menu is open, showing 'day_of_week' and 'WeekdayNumeric'. The 'WeekdayNumeric' option is selected, indicating that the data is sorted by this column.

Count of guests booking by day_of_week and hotel



Regarding when "reservation_status_date" & "arrival_DD_MM_YYYY" is used during visualization

Field Name	Description
reservation_status	Reservation last status, assuming one of three categories: Canceled – booking was canceled by the customer; Check-Out – customer has checked in but already departed; No-Show – customer did not check-in and did inform the hotel of the reason why
Reservation_status_date	Date at which the last status was set. This variable can be used in conjunction with the ReservationStatus to understand when was the booking canceled or when did the customer checked-out of the hotel



- If we want to answer revenue-related questions with plots, we'll use "reservation_status_date" as the x-axis. This is because as defined in our SQL query, revenue is only generated when reservation_status is either "Check-Out" or "No-Show" having an adr >= 0.
 - For a revenue generating entry, since the "reservation_status_date" date value will always update to the latest date in which the customer has "check-out" or "No-Show", it is a more accurate representation of when revenue is generated as compared to using "arrival_DD_MM_YYYY" as the x-axis
 - Conversely, this also assumes that the hotel's revenue is generated when the guests physically check-out from the hotel itself
 - This also means that we assume that all revenue in this data analysis is not generated from online booking

Field Name	Description
Arrival_date_year	Year of arrival date
Arrival_date_month	Month of arrival date
Arrival_date_week_number	Week number of year for arrival date
Arrival_date_day_of_month	Day of arrival date

- Conversely, if we want to answer questions such as "Which are the most busy month?" & parking-related questions, We'll use "arrival_DD_MM_YYYY" column as it'll be a more accurate time representation of when the guests have physically "checked-in" into the hotel

Elaborating on 'reservation_status_date'

- For “Check-Out”, we see that “reservation_status_date” is $\geq$ arrival date.
 - This means that the date listed in reservation_status_date is updated at the same time when the customer has checked out and thus the revenue is generated
- For “No-Show”, we see that that “reservation_status_date” = arrival date

Takeaway reminder:

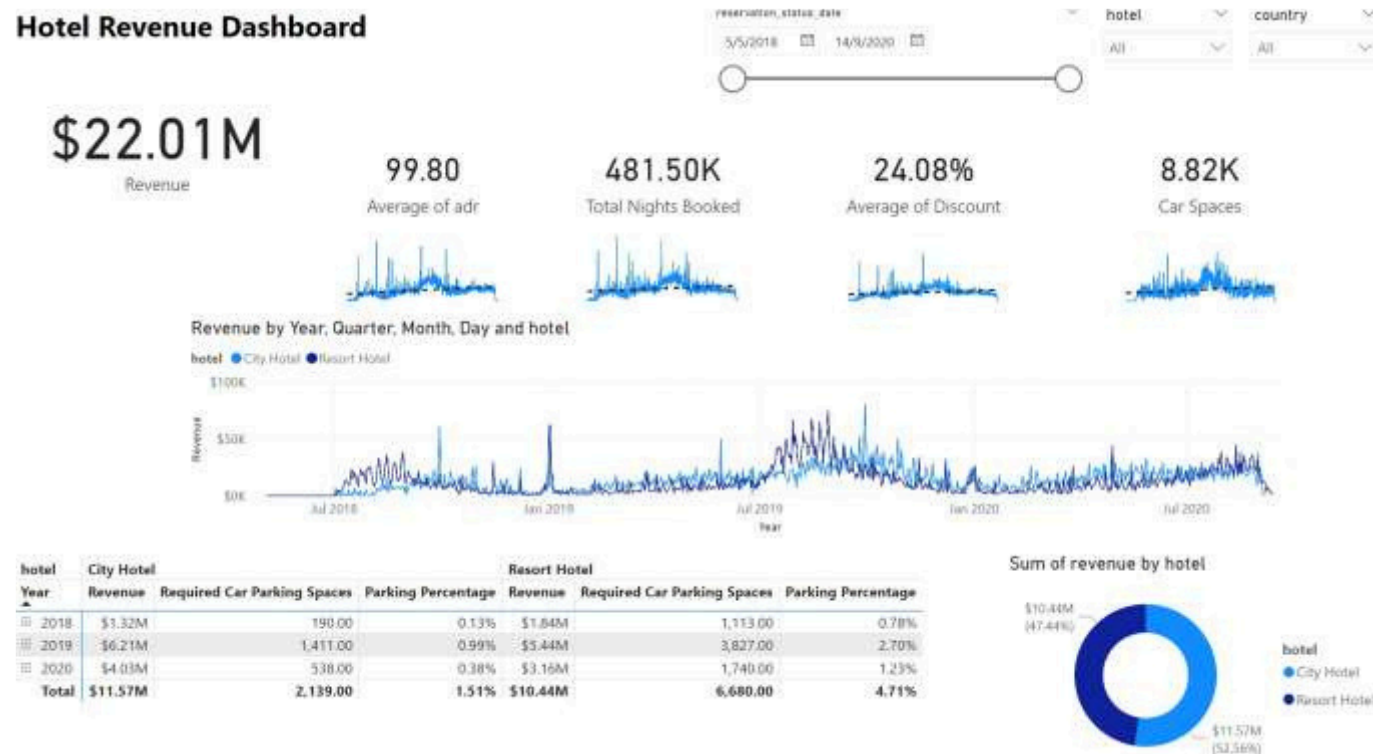
Reservation_status_date IS NOT THE DATE where the customer has reserved the booking prior to arriving at the hotel

reservation_status	reservation_status_date	arrival_Concat	revenue
Check-Out	12/9/2019	5July2019	6813.01
Check-Out	30/9/2018	1August2018	5949.01
Check-Out	30/9/2019	1August2019	5949.01
Check-Out	29/8/2020	15August2020	3993.81
Check-Out	14/8/2018	1August2018	3462.76
Check-Out	14/8/2019	1August2019	3462.76
Check-Out	11/11/2019	19October2019	3310.51
Check-Out	13/8/2020	3August2020	3294.91
Check-Out	20/8/2020	6August2020	3150.028
Check-Out	27/6/2020	16June2020	3134.328
Check-Out	13/7/2019	2July2019	3116.31
Check-Out	31/7/2020	1July2020	3087.01
Check-Out	9/8/2020	1August2020	3069

reservation_status	reservation_status_date	arrival_Concat	revenue
No-Show	30/6/2019	30June2019	3066.828
No-Show	2/12/2018	2December2018	2870.61
No-Show	2/12/2019	2December2019	2870.61
No-Show	12/5/2019	12May2019	2286.465
No-Show	3/2/2019	3February2019	2164.794
No-Show	13/2/2019	13February2019	2085.624
No-Show	9/5/2019	9May2019	1985.955
No-Show	3/9/2019	3September2019	1916.728
No-Show	12/1/2019	12January2019	1902.21
No-Show	30/3/2019	30March2019	1869.492
No-Show	30/6/2019	30June2019	1851.684
No-Show	25/3/2019	25March2019	1773.6
No-Show	22/11/2018	22November2018	1744.899
No-Show	22/11/2019	22November2019	1744.899
No-Show	15/3/2020	15March2020	1700.61
No-Show	4/7/2019	4July2019	1514.97
No-Show	3/3/2019	3March2019	1510.532
No-Show	12/2/2019	12February2019	1503.91
No-Show	25/2/2019	25February2019	1498.835
No-Show	17/4/2020	17April2020	1485.51

Hotel Revenue Dashboard

Hotel Revenue Dashboard



It provides at-a-glance information about features that are relevant to revenue generation

It also provides a comparison of revenue generated by different hotels, and parking space utilization information

Hotel guest-related seasonality EDA



This section will mainly focus on relevant visualizations that will aid in answering the following questions:

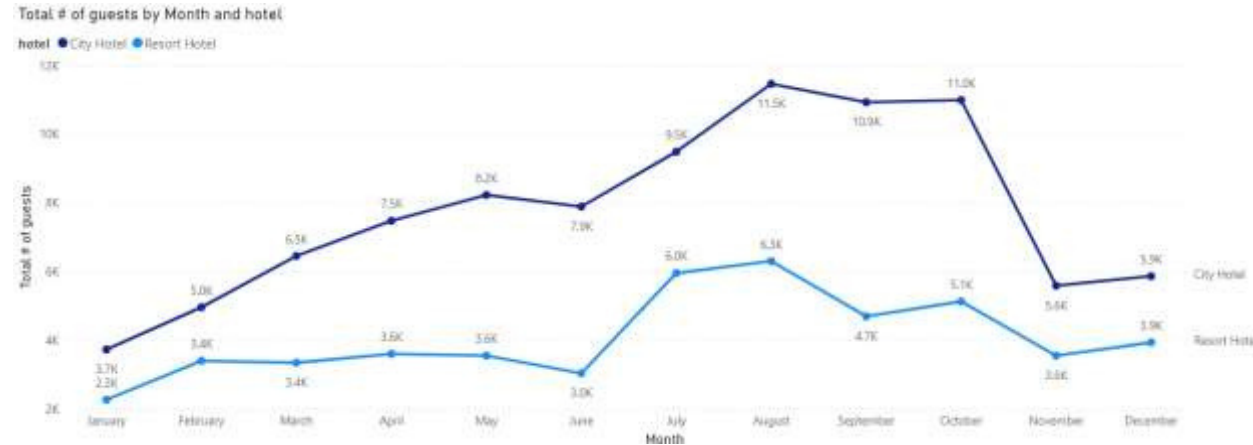
- “Which are the busiest months?”
- “On average, how long do guests stay at the hotel?”
- “Which day of the week do different types of guests, typically like to book on?”

Breakdown on the type of guests that are booking the hotel will be provided

Relevant recommendations will be provided for the hotel manager to improve staffing allocation based on guest booking arrival data

Hotel guest-related seasonality EDA

“Which are the busiest months?”



- From the line chart, we observe that the busiest months across all years are:
 - July & August for Resort Hotels
 - July to October for City Hotels

Recommendation: The hotel manager should take note to plan service staff shift accordingly to allocate more hotel staffing during those critical high volume months

On average, how long do guests stay at the hotel?

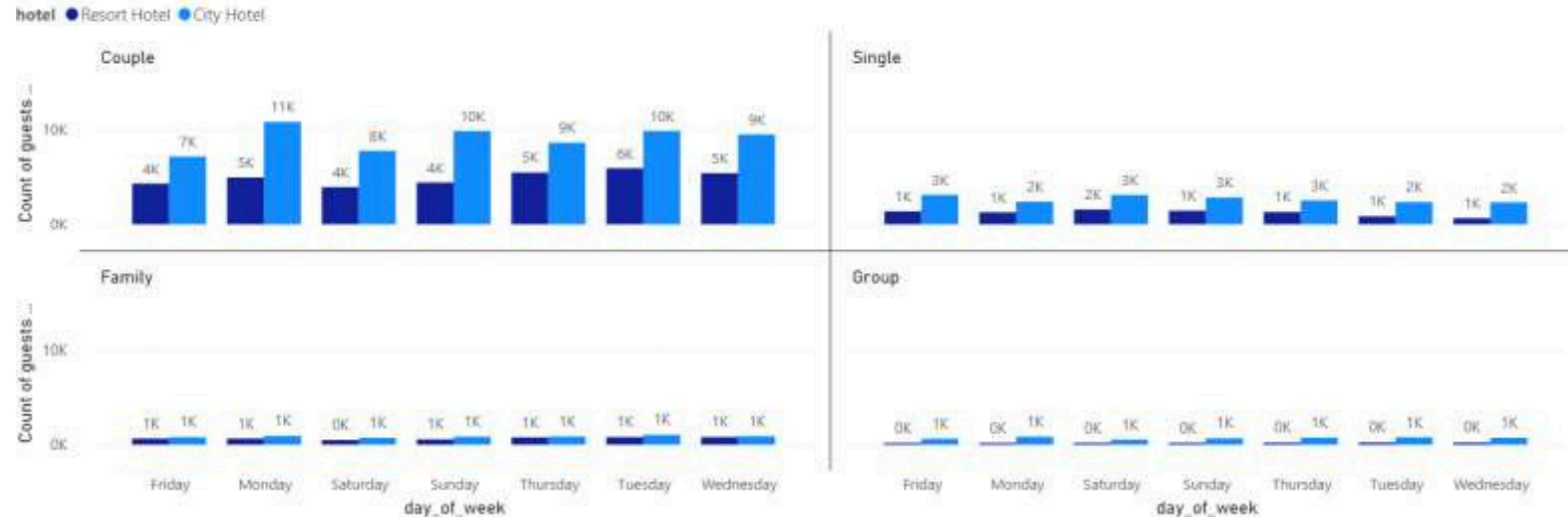


- On average, guests stay **about 3.42 nights** for both hotels

Hotel guest-related seasonality EDA

“Which day of the week do different types of guests, typically like to book on?”

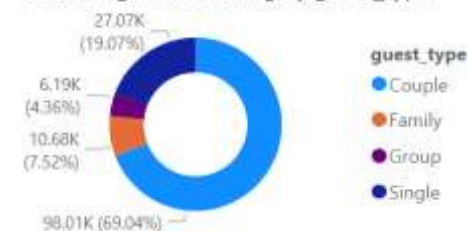
Count of guests booking by day_of_week, hotel and guest_type



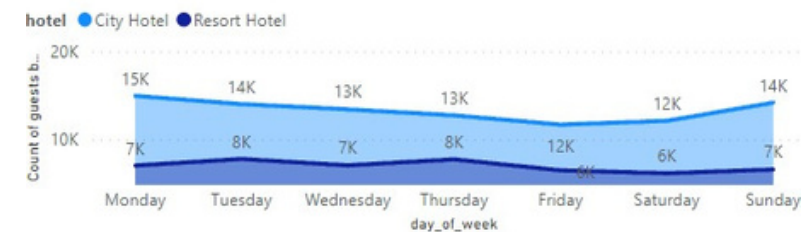
Based on the data, we infer that:

- Couples and single guests typically account for the majority of bookings made for both hotels
- There are significantly more bookings made by Couple & Single guests for City Hotels as compared to Resort Hotels
- The number of City Hotel bookings made by group guests are about ten-fold for every day of the week as compared to Resort Hotels

Count of guests booking by guest_type



Count of guests booking by day_of_week and hotel



- For both hotels, the count of guests arriving for each day of the week appears to be quite evenly distributed

Recommendation: Management could look into designing attractive packages tailored towards Couple & Single guests for both hotels, given that they represent the majority of the customers

Hotel Revenue EDA



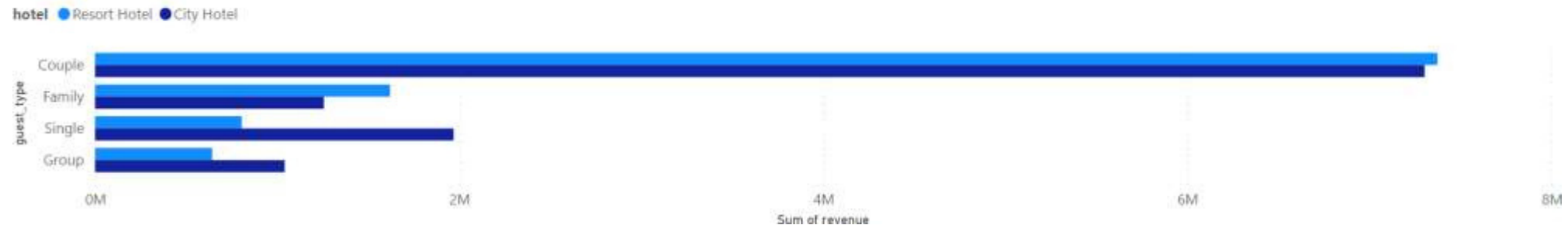
This section will mainly focus on relevant visualizations that will aid in answering the following questions:

- “Which type of guests are likely to generate more revenue by hotel?”
- “Give a revenue breakdown by region. What are the top 5 countries that have generated the greatest revenue?”
- “Is our hotel revenue growing by year? If so, which hotel brings in greater revenue? Is there any observed seasonality with respect to revenue?”

Hotel Revenue EDA

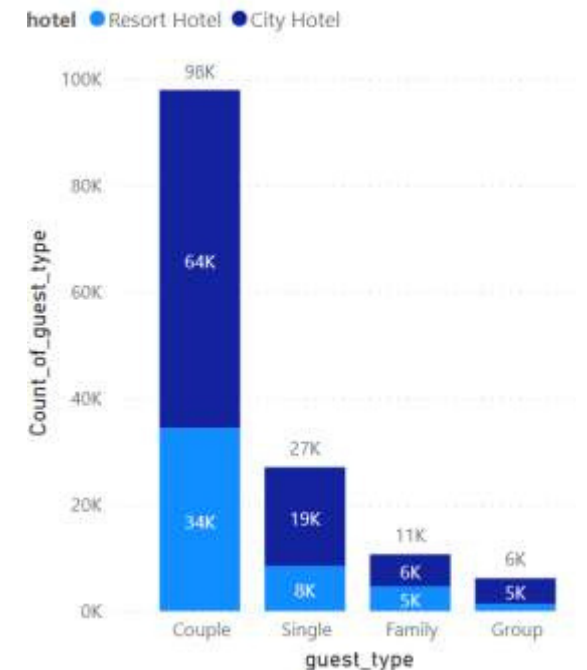
“Which type of guests are likely to generate more revenue by hotel?”

Sum of revenue and Count_of_guest_type by guest_type and hotel



- **Couples and family guests tend to generate more revenue in resort hotels as compared to city hotels. While single and group guests tend to generate more revenue in city hotels.**
- The revenue difference is marginal between hotels for couples. This is in spite of there being almost twice the number of couples who booked for city hotel (as indicated in the stacked bar plot to the right)
 - This marginal difference in revenue despite fewer bookings in resort hotels may be due to the presence of more expensive amenities (such as pools & spa services) packaged for couples and families in resort hotels
- On the other hand, **more revenue has been generated in city hotels for single and group hotel guests**
 - This is likely attributed by city hotel having more bookings for single and group guests as compared to resort hotels
 - This is in-line with what was previously mentioned, that couples and families are likely to have bundled extra amenities when they had booked for resort hotels hence attributing to a higher revenue despite fewer bookings

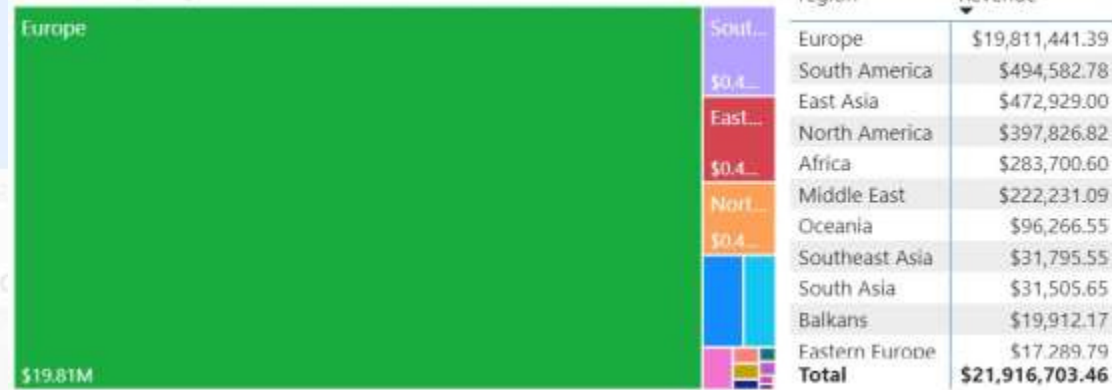
Count_of_guest_type by guest_type and hotel



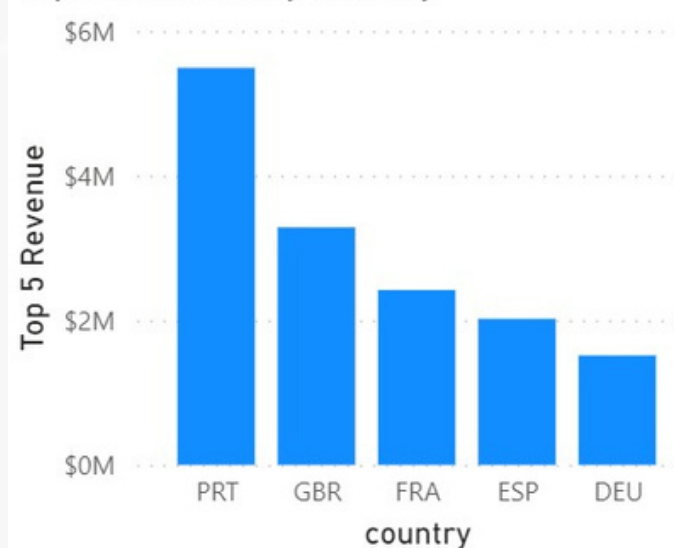
Hotel Revenue EDA

“Give a revenue breakdown by region. What are the top 5 countries that has generated the greatest revenue?”

Revenue by region



Top 5 Revenue by country



- The main bulk of revenue generated were from guests in the European region from 2018-2020

- The **corresponding top 5 countries** by revenue are:

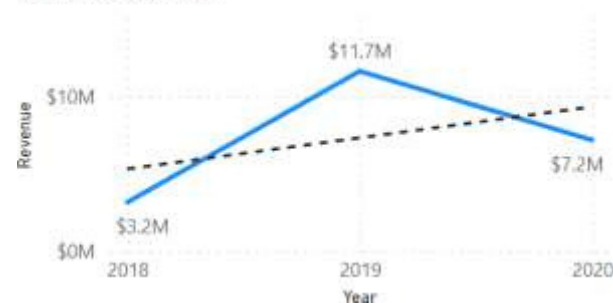
1. PRT – Portugal, \$5.49M
2. GBR – United Kingdom, \$3.28M
3. FRA – France, \$2.42M
4. ESP – Spain, \$2.02M
5. DEU – Germany, \$1.51M

The top 5 countries by revenue are all situated in the European region.
This may suggest that the location of the hotels may likely to be situated in the European region

Hotel Revenue EDA

“Is our hotel revenue growing by year? If so, which hotel brings in greater revenue? Is there any observed seasonality with respect to revenue?”

Revenue by Year



- Yes, our overall hotel revenue is growing by year and there is an upwards growth trend as depicted in the trendline
 - Revenue trended up, resulting in a **127.20% increase** between 2018 and 2020.
 - 2019 had the **highest combined revenue recorded at \$11.7M**, which is 254.18% higher than the lowest revenue recorded at \$3.2M
- From 2019 to 2020, we see a decrease in revenue from \$11.7M to \$7.2M.
 - This may be due to the influence of COVID-19 pandemic which may have impacted the hotel industry [The first recorded case of COVID-19 death in Portugal was recorded on 16 MAR 2020]

Sum of revenue by hotel



- City Hotel brings in greater revenue [52.56% of total revenue] as compared to Resort Hotel [47.44% of total revenue]

Revenue by Quarter and hotel

hotel ● City Hotel ● Resort Hotel



Revenue by Month and hotel

hotel ● City Hotel ● Resort Hotel



Seasonality trend based on limited dataset from 2018-2020

- Across all years, it was observed that revenue typically picks up from the end of Q2 (June), cumulating in a maximum in the month of August

Limitations of dataset

Limited data on 2018 & 2020 may paint an incorrect conclusion for revenue & guest-related seasonality reporting

- More complete dataset containing information from **2018 Q1, Q2 & 2020 Q4** is required for the seasonality analysis to be more accurate

Year	City Hotel	Resort Hotel	Total
2018	\$1,323,071.40	\$1,843,214.99	\$3,166,286.40
Qtr 2	\$0.00	\$0.00	\$0.00
Qtr 3	\$673,004.79	\$1,405,697.04	\$2,078,701.83
Qtr 4	\$650,066.62	\$437,517.95	\$1,087,584.57
2019	\$6,212,979.87	\$5,440,438.45	\$11,653,418.32
Qtr 1	\$824,685.11	\$482,909.39	\$1,307,594.50
Qtr 2	\$1,326,433.70	\$809,577.56	\$2,136,011.26
Qtr 3	\$2,299,979.11	\$3,127,562.34	\$5,427,541.45
Qtr 4	\$1,761,881.95	\$1,020,389.16	\$2,782,271.11
2020	\$4,034,910.86	\$3,158,776.11	\$7,193,686.96
Qtr 1	\$985,558.64	\$506,782.76	\$1,492,341.40
Qtr 2	\$1,747,040.20	\$1,030,665.24	\$2,777,705.44
Qtr 3	\$1,302,312.02	\$1,621,328.11	\$2,923,640.13
Total	\$11,570,962.13	\$10,442,429.55	\$22,013,391.68

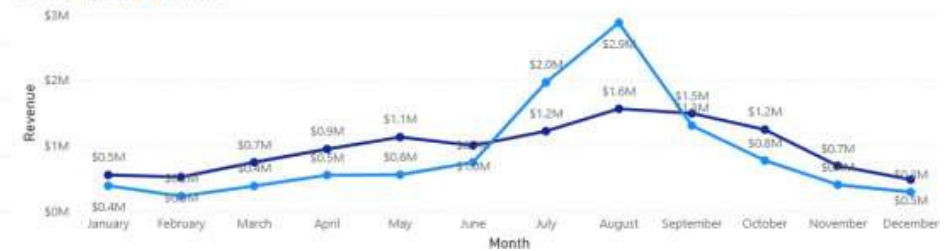
Revenue by Quarter and hotel

hotel ● City Hotel ● Resort Hotel



Revenue by Month and hotel

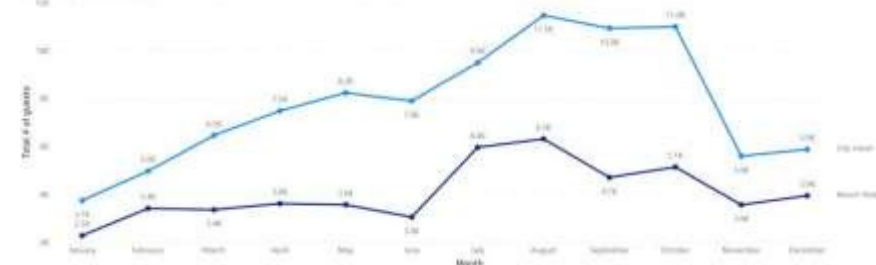
hotel ● City Hotel ● Resort Hotel



Year	City Hotel	Resort Hotel	Total
2018	13682	8314	21996
2019	51913	27351	79264
2020	27508	13179	40687
Total	93103	48844	141947

Total # of guests by Month and hotel

hotel ● City Hotel ● Resort Hotel



Limitations of dataset

More information about car parking spaces is required to be provided into the dataset to accurately calculate the parking percentage

- For example, a more accurate representation of the calculated 'Parking Percentage' should be obtained with a column containing 'total_car_parking_spaces_available', where:

- $$\text{ParkingPercentage} = \frac{\text{Required Car Parking Spaces}}{\text{Total Car Parking Spaces Available}} \times 100\%$$

Field Name	Description
Required_car_parking_spaces	Number of car parking spaces required by the customer

hotel	City Hotel			Resort Hotel		
Year	Revenue	Required Car Parking Spaces	Parking Percentage	Revenue	Required Car Parking Spaces	Parking Percentage
2018	\$1.32M	190.00	0.13%	\$1.84M	1,113.00	0.78%
2019	\$6.21M	1,411.00	0.99%	\$5.44M	3,827.00	2.70%
2020	\$4.03M	538.00	0.38%	\$3.16M	1,740.00	1.23%
Total	\$11.57M	2,139.00	1.51%	\$10.44M	6,680.00	4.71%

Currently, the Required Car Parking Spaces in the dashboard is calculated as such:

```
1 Parking Percentage = SUM('hotel_all_column_amalgated with 5 categorical columns'[required_car_parking_spaces])/SUM('Count of Reservation Status Made'[Count of Reservations made])
```

- $$\text{Parking Percentage} = \frac{\text{Required Car Parking Spaces}}{\text{Count of Reservation Status Made}} \times 100\%$$

The validity of the calculation can be further improved if we can obtain a 'total_car_parking_spaces_available' column for management to decide if a larger parking lot is required to be built.

Conclusion / Key-Takeaways

Based on the limited dataset provided, below are the key-takeaways from this project:

With respect to hotel guest-related seasonality:

- The hotel manager should ensure increased hotel staffing during these months for each hotel:
 - o July & August for Resort Hotels
 - o July to October for City Hotels
- Guests on average stay 3.42 nights for both hotels
- No particular trends can be observed as to which days are more popular for guests to book on
- Couples & single guests represent the majority of bookings made for both hotels, management can look into designing attractive packages tailored towards such guests

With respect to Hotel Revenue:

- Couples & Family guests tend to generate more revenue in resort hotels as compared to city hotels, while group guests generate more revenue in city hotels
- The main bulk of revenue generated was attributed to the European region
 - Portugal corresponded to the country with the highest revenue
- Hotel revenue displays an upwards trend and is growing by year
- City Hotel brings in greater revenue as compared to Resort Hotel
- Revenue typically starts to pick up from the end of Q2 (June), cumulating in a maximum in the month of August

END OF
PRESENTATION